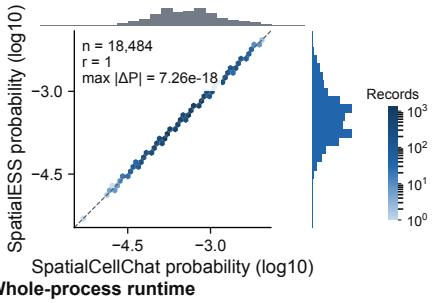
B Probability fidelity · CosMx 50k

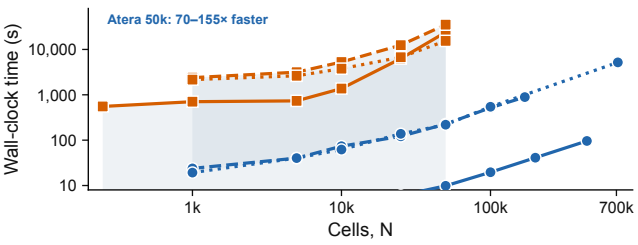


C Fidelity across paired scales

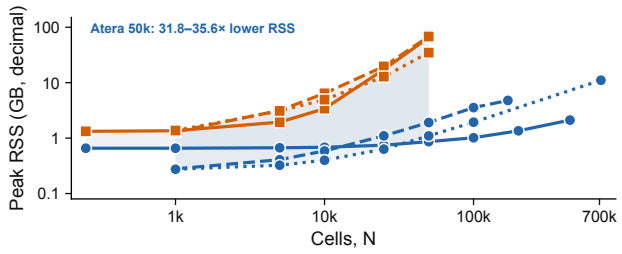
	250	1k	5k	10k	25k	50k
Probability r	1	1	1	1	1	1
Max |Δ probability|	5.96e-19	1.73e-18	2.60e-18	3.69e-18	7.81e-18	7.26e-18
Exact p fraction	1	1	1	1	1	1
Active-record Jaccard	1	1	1	1	1	1
Call disagreements	**0**	**0**	**0**	**0**	**0**	**0**

Atera 1–50k: 10/10 strict passes; max |ΔP| = 2.98e-16; p identical; 0 call disagreements

D Whole-process runtime



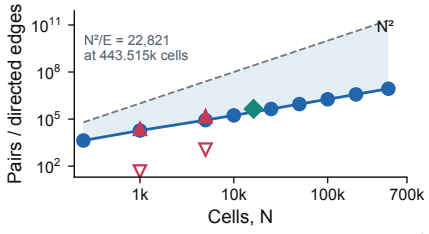
E Peak resident memory



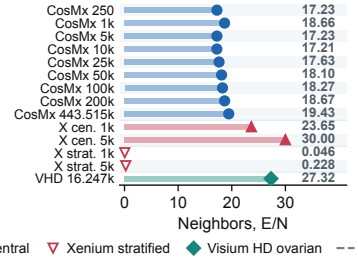
SpatialESS SpatialCellChat

CosMx CancerousLiver Atera breast Atera cervical Shading: paired completed sizes

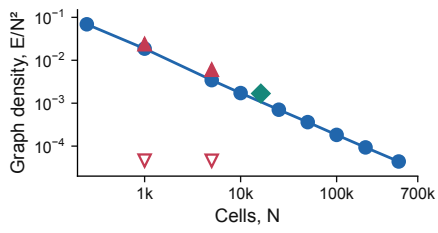
F Potential pairs and observed edges



G Local neighbors per cell



H Graph density



CosMx Xenium central Xenium stratified Visium HD ovarian N²: all possible ordered pairs

Directed non-self edges; effective radius = 35 µm. Runtime and memory points are measured values.

I Completion and comparison scope

